ORIGINAL RESEARCH

# Comparative Analysis of CNN Models for Amaranthus Leaf Disease Detection

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**Abstract** Amaranthus is a crop with high nutritional value that is widely grown in tropical and subtropical areas around the world. Despite its significance, the crop faces major yield losses each year due to various leaf diseases, causing losses of about 30 to 50 percent. Traditional methods for detecting these diseases often take a lot of time, are subjective, and need expert knowledge, which makes them hard for small-scale farmers to use. To solve these issues, new advancements in artificial intelligence and image processing have led to developing automated, efficient, and objective systems for diagnosing plant diseases.

This document presents an intelligent deep learning-based system that automatically detects Amaranthus leaf diseases through image analysis. The system uses convolutional neural networks (CNNs) and employs transfer learning to evaluate and compare the performance of three top architectures: MobileNetV2, ResNet50V2, and DenseNet121, for classifying multiple diseases. A detailed dataset containing labeled images of healthy leaves and five major Amaranthus diseases, gathered from both fields and laboratories, was used for training and validating the model. Standard preprocessing, data augmentation, and fine-tuning techniques were used to improve the model's strength and adaptability. The experimental results show that ResNet50V2 reached the highest validation accuracy among the models tested, while MobileNetV2 showed strong performance with significantly lower computational needs, making it a better choice for real-time use in low-resource settings. The system is integrated into an easy-to-use web platform that allows farmers and agricultural professionals to upload leaf images and quickly receive disease predictions along with confidence scores. This method helps with early disease detection, supports timely treatment decisions, and encourages better crop management and sustainable farming practices.

**Keywords** Amaranthus crops · Leaf disease detection · Deep learning · Convolutional neural networks · Precision agriculture

## 1 Introduction

Amaranthus is commonly cultivated as a leafy vegetable and grain. Its crucial role in food security stems from its high nutritional content and its ability to adapt to diverse climates. Nevertheless, Amaranthus is highly vulnerable to various leaf afflictions, including white rust, stem canker, wet rot, damping off, and other fungal infections.

The conventional techniques for plant disease diagnosis have involved the use of expertise from professional horticulturists who undertake direct analysis of the plants. This technique is time-consuming, requires human labor, and is associated with human error, making it unsuitable for use in rural settings. Thus, there is a high need for a technique that can efficiently and accurately diagnose diseases in the Amaranthus species.

Computer vision and deep learning techniques have improved the diagnosis of plant diseases in agriculture by enabling the automatic detection of diseases from digital images of leaves. Convolutional Neural Networks (CNNs) have been shown to be more accurate in image classification tasks, including the classification of plant diseases from leaf images. However, most of the literature has been dominated by plants that are common in agriculture, such as tomatoes, potatoes, and rice, while leafy vegetables like Amaranthus have been less

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represented in the literature on plant disease identification. In addition, the models that have been developed have been trained on data that has been collected in a controlled lab environment, which may not be representative of the actual environment in agriculture.

Recent research has proved the efficacy of CNN-based models for plant disease diagnosis through leaf image analysis. Transfer learning has been a significant boost to classification accuracy, especially in the case of a limited number of samples. Pre-trained models such as ResNet50V2, DenseNet121, and MobileNetV2 have proved to be effective for feature learning and transfer. However, there has been less emphasis on leafy vegetables, especially Amaranthus, in image analysis conditions with noise.

In this regard, the aim of the current study is to provide a deep learning solution that is suitable for the detection of leaf diseases in the Amaranthus species. By applying CNN models and transfer learning, the proposed solution aims to provide a feasible, effective, and accurate method for farmers to obtain real-time diagnoses of the diseases. The proposed solution not only aims to improve the accuracy and efficiency of the diagnosis but also aims to fill the gap between research and application.

Amaranthus is a major source of nutritional security, especially in developing nations where protein deficiency is a widespread problem. The leaves and seeds of the plant are consumed by people as they are rich in nutrients, such as lysine, dietary fibers, and calcium and iron. Moreover, this plant can grow well in different conditions, even in drought and poor soil. But the productivity of Amaranthus is affected by various biological factors, which are mainly due to fungal and bacterial infections in the leaves, stems, and roots. Some of the major diseases that affect Amaranthus are white rust (Albugo bliti), damping off (Pythium spp.), and stem canker (Phoma spp.). These diseases result in the destruction of the entire crop, which is a major problem for farmers in the region.

The traditional method of inspecting crops manually for disease diagnosis has several limitations. To make an accurate diagnosis, one requires pathologists who are trained and have been monitoring the fields regularly. Moreover, the process of sampling requires control, which may not be possible in large fields. In addition to this, the diagnosis may be subjective, depending on the symptoms observed. Therefore, the use of artificial intelligence in plant disease management systems is a more efficient alternative to the traditional method.

Currently, computer vision and deep learning are emerging technologies in the field of precision agriculture. Machine learning algorithms, particularly Convolutional Neural Networks (CNNs), have revolutionized image analysis by being able to detect features automatically without the need for human intervention. The algorithms are able to detect complex patterns, textures, and variations in leaf color in images, which may not be easily visible to the human eye. This has resulted in a rise in the use of AI in agriculture.

Despite this progress, a significant challenge remains: most studies focused on deep learning target major crops like maize, wheat, tomatoes, and potatoes, while neglected but nutritious crops like Amaranthus receive little research attention. Also, the datasets used to train these models often come from controlled lab settings (e.g., the PlantVillage dataset), where lighting and backgrounds are kept constant. While these datasets help in model development, they don’t capture the real-life complexities of farms, where leaves may overlap, change orientation, or be partly hidden by environmental factors like soil or moisture. This gap lowers model reliability and complicates field deployment.

The adoption of transfer learning has been effective in handling the limited data available in the agricultural sector. This is because transfer learning enables the use of pre-trained models that were trained on large datasets such as ImageNet. This increases the speed of training and the efficiency of feature extraction from a small dataset. Some of the models that have been effective include ResNet50V2, DenseNet121, and MobileNetV2. These models are a combination of accuracy and efficiency, making them the best for the development of mobile and web applications that farmers can access.

However, there is still a need for appropriate frameworks to automatically identify Amaranthus leaf diseases from the leaves of Amaranthus. The previous models are likely to perform poorly when applied to natural images captured in different settings. This is because the images are likely to have varying lighting conditions, backgrounds, and leaf textures. This is a clear indication that there is a need to develop models that can adapt to real-world settings. This study will therefore help to solve the problem by developing a system for Amaranthus leaf disease detection using deep learning that combines image preprocessing, feature extraction, and classification. The system will be trained on a selected dataset that includes various types of diseases of Amaranthus to ensure diversity in images and equality in classes. In addition, the system will also utilize transfer learning with CNN models such as ResNet50V2, MobileNetV2, and DenseNet121 to ensure that the system is both accurate and efficient, allowing it to be applied on devices with limited capabilities, such as smartphones and tablets, which will be very useful for small-scale farmers who can utilize the system to detect diseases in real-time by taking leaf photos of the Amaranthus plant in the field.

## 2 Literature Review

Automated plant disease detection using deep learning has become more popular in recent years. Mohanty et al. demonstrated that CNN architectures like AlexNet and GoogLeNet can effectively classify plant diseases with the PlantVillage dataset, achieving high accuracy in controlled settings. Ferentinos built on this by exploring deep CNN models and found strong performance across various plant species, though these models require substantial computational power. Sladojevic et al. applied deep CNNs to custom datasets and obtained promising results, but their findings were limited by the size of the datasets. Research on using deep learning for automatic plant disease recognition has increased due to its potential to improve agricultural efficiency and reduce reliance on human assessments [1]. Earlier research by Mohanty et al. demonstrated the effectiveness of convolutional neural network (CNN) models like AlexNet and GoogLeNet in identifying plant diseases through the PlantVillage dataset, with a focus on laboratory settings [2]. Their work showed that deep learning models can automatically spot unique visual features in leaf images without needing manual feature creation. Building on this foundation, Ferentinos explored more complex CNN designs and achieved better classification accuracy across various crop species and disease types [3]. However, the high computational requirements of these models limited their practical use on devices with fewer resources. Similarly, Sladojevic et al. had encouraging results with their deep CNN models oncustom plant disease datasets, but faced challenges due to small dataset sizes and inconsistent image collection conditions [4].

To address the problem of limited labeled training data, many research projects have turned to transfer learning methods, which use pre-trained models to boost accuracy and speed up training [5]. Brahimi et al. employed transfer learning techniques for classifying tomato leaf diseases, achieving high accuracy by fine-tuning existing CNN models [6]. While these results are promising, their approach focused only on a single crop species and did not explore cross-crop generalization. Recent studies have investigated more advanced network designs like DenseNet and ResNet, which showed better accuracy and faster learning rates on standard benchmarks [7,8]. These designs use dense and residual connections to improve feature reuse and ensure stable training efficiency. Despite most evaluations relying on data from controlled environments, there are concerns about the model's performance in real-world agricultural settings [9,10,11,12].

Research using field-collected images has revealed additional challenges affecting classification accuracy, such as complex backgrounds, changing lighting conditions, shadows, and partial leaf occlusion [12]. As a result, models trained and tested in controlled locations often perform worse when applied in real-world situations. Additionally, many existing deep learning-based disease detection systems use resource-heavy models, making them unsuitable for low-power devices like smartphones and edge computing platforms commonly used in rural agriculture [10]. Moreover, there is a notable lack of studies focused on leafy vegetables like Amaranthus, despite their economic and nutritional importance. This research introduces a disease detection system for Amaranthus leaves that leverages transfer learning and incorporates images taken in real-world conditions [13]. The goal is to improve classification accuracy while ensuring computational efficiency, making it practical for use in real-world scenarios.

Further studies have examined ways to optimize deep learning models for deployment on resource-limited devices. Picon et al. analyzed the use of CNNs for crop disease classification on mobile capture devices and demonstrated that lightweight models can achieve reasonable accuracy while remaining practical for field use [11]. Similarly, Too et al. conducted a comparative study on fine-tuning pre-trained deep learning models for plant disease identification. They found that choosing pre-trained networks and adjusting hyperparameters greatly affects performance, especially with limited dataset sizes [9]. These findings highlight the need to balance model complexity with computational efficiency for real-world applications [10].

Barbedo studied various factors influencing deep learning-based plant disease recognition, including image quality, dataset size, and variability in environmental conditions [10]. He observed that these factors often limit model performance outside controlled lab settings. Ferentinos and Alifieris provided a thorough review of deep learning models for plant disease detection [12]. They noted that while CNN architectures have achieved high accuracy across diverse crop types, there is still a need for models that can handle cross-crop situations and different field conditions [12]. These insights motivate the current study to focus specifically on Amaranthus crops, which have received limited attention in prior research, and to use transfer learning to achieve both high accuracy and computational efficiency in field settings [13].

## 3 Proposed Model

The Fig.1 above illustrates the complete system setup for the prediction of the Amaranthus leaf disease using deep learning. The system begins with data collection, which entails the use of both artificially created images and actual images of healthy and diseased Amaranthus leaves.

The preprocessing and augmentation stage of the system is responsible for resizing the images to the same size and adjusting the brightness of the images. Moving forward, the deep learning process employs three CNN models: MobileNetV2, ResNet50V2, and DenseNet121. These models are trained and validated on a chosen dataset through transfer learning. The models attain validation accuracies of 97.0%, 99.57%, and 99.35%, respectively. The most effective model is then incorporated into a web-based deployment environment, which consists of a frontend user interface, a backend server, and a CNN model integration module.

Finally, the user interface of the system enables farmers to upload images of leaves from their smartphones or computers for rapid identification of diseases and treatment. This helps to bridge the gap between research in AI and its application in agriculture.

Fig 1. Workflow of the Proposed Deep Learning Framework for Amaranthus Leaf Disease Prediction

### 3.1 Data Collection

The dataset for this research was carefully created to support effective and relevant detection of Amaranthus leaf diseases. We acquired images using a mix of field photography and controlled lab imaging. This ensured we captured environmental variety and visual clarity.

Field images were taken from local farms and experimental plots with high-resolution smartphone cameras in natural daylight. This method introduced realistic changes, like variations in lighting, leaf position, size, and background complexity—conditions we typically see in actual farming settings.[16]

We also obtained laboratory images from Amaranthus samples showing disease symptoms under controlled lighting and simple backgrounds.[15] This allowed us to clearly capture the features of each disease. These lab samples provided reliable benchmarks for identifying disease symptoms and improving how the model interprets results.

To improve dataset diversity and tackle class imbalance, we generated some synthetic samples using AI data augmentation tools. This ensured that less common disease classes were well represented. The final dataset included five main leaf categories: Healthy, White Rust, Stem Canker (Anthracnose), Wet Rot (Stem Rot), and Damping Off (Root Rot).[14]

Agricultural experts manually verified and annotated all samples to guarantee high-quality labeling and consistency across categories. This combined strategy—merging real-world variability, lab precision, and expert verification—helped create a strong and representative dataset for deep learning-based disease prediction.

This data collection method follows established practices in plant disease detection studies, like those from Mohanty et al. (2016) and Ferentinos (2018), which highlight the importance of using varied imaging sources and expert annotation for better model performance.

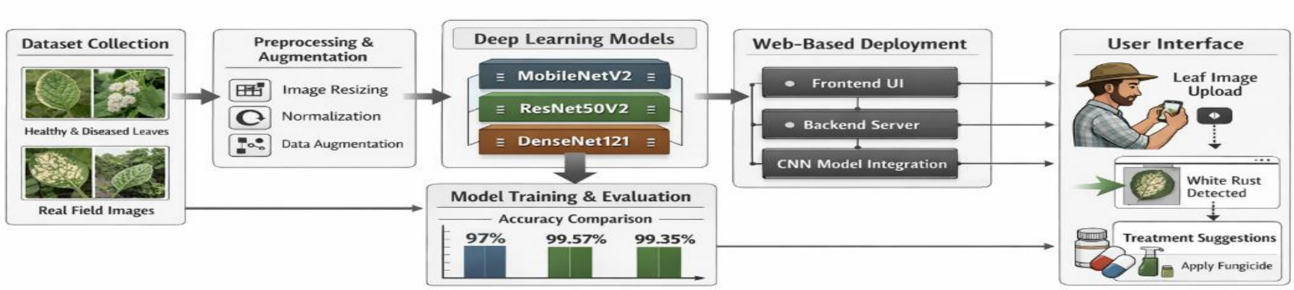

*Table 1: Distribution of images per class used in the dataset.*

| Class | Total Images | Laboratory Images | Field Images |
|---|---|---|---|
| **Healthy** | 1200 | 100 | 63 |
| **White Rust** | 1150 | 100 | 59 |
| **Stem Canker** | 1180 | 100 | 75 |
| **Wet Rot** | 1142 | 100 | 71 |
| **Damping Off** | 1140 | 100 | 67 |

*Figure 2 Sample Image-1*

## 3.2 Data Preprocessing and Augmentation

To ensure the robustness and generalization of the deep learning models, a systematic preprocessing and augmentation pipeline was implemented. The objective was to standardize the input images, reduce noise, and artificially expand the dataset through controlled transformations that simulate real-field variations.

### 3.2.1 Image Standardization and Enhancment

Every retrieved image is resized to the fixed dimension 224 × 224 × 3, which is essential in satisfying architectural input constraints of CNN model variants such as MobileNetV2, ResNet50V2, and DenseNet121. The pixel intensity values are then normalized between 0 and 1 by dividing each pixel value by 255. This normalization ensures numerical stability while training and provides consistent scaling across all the input images.

Images acquired in the real field normally possess variations in illumination and can exhibit some noising artifacts. In that aspect, an enhancement scheme for selected images has been carried out. Those images which revealed extreme glare or motion blur were not considered in the dataset. Adaptive histogram equalization has been performed on those images showing minor inconsistencies in lighting to enhance the contrast for highlighting the disease-affected regions. Enhancement thereby increases the visibility of relevant patterns and enhances the effective extraction of discriminative spatial features by convolutional layers.

### 3.2.2 Normalization

Normalization was done to rescale all image pixel intensities from their original range, [0, 255], to the range [0, 1], for the purpose of numerical stability and ensuring that features are identically distributed across channels.
This scaling avoids large gradient fluctuations during training for smoother and faster model convergence.
By reducing the internal covariate shift, it stabilizes the layer's activations and improves the generalization capability of deep CNN architectures.
In addition, it blocks dominance of high-intensity features, enabling equal learning contribution from all pixels.
Thus, the models effectively detected fine-grained features related to disease manifestations, such as rust spots and leaf lesions, for a wide range of illumination conditions.

### 3.2.2 Noise Reduction and Enhancement

Variations in noise, shadow, and illumination that can obscure image clarity may be observed in the captured Amaranthus leaf images. In response to this issue, a Gaussian filter was employed to eliminate noise present in the higher-frequency components. Bear in mind that, according to theory, this method might result in data loss if not executed correctly. These steps were followed by applying an adaptive contrast to improve the visibility of disease-specific areas. Both approaches enhanced the image quality, thereby facilitating accurate feature extraction by CNNs. Thus, this resulted in the network demonstrating improved ability to detect subtle disease symptoms across various environmental conditions.

### 3.2.3 Data Augmentation Strategy

To overcome the limitations of the size of the datasets and also avoid overfitting, data augmentation was used dynamically during the training phase. A set of random geometric and photometric transformations were employed, simulating real variations as in agricultural crop fields. The transformations used included rotation of images within ±25°, flip of images horizontally and vertically, zoom functions with a scaling limit of 15%, shear transformation functions with limits of ±20°, and intensity functions within ±20%. Additionally, there was random cropping, used to draw attention to the regions of interest related to plant disease symptoms.

All augmentation tasks were performed in a probability-driven fashion by employing the Keras Image Data Generator and the Albumentations library to ensure a different set of samples is presented to the network for every epoch during training. This helped create variability in the dataset while retaining characteristics of interest that are specific to a particular disease.

### 3.2.4 Impact on Model Generalization

It enabled this pipeline of preprocessing and augmentation to offer a standardized yet diversified training dataset with capabilities to tackle various real-world settings like background mess, illumination, and orientation. The variability was managed via preprocessing, which ensured that the models retained their distinctive visual features and showed improved convergence performance with reduced overlap.

## 3.3 Data Splitting

In order to have a non-biased test on models, data was split into training and test data with a stratified structure that is composed of 80% and 20%, respectively. Data stratification ensured that it is evenly represented for all six classes.

| Data Split | Number of Images | Percentage | Purpose |
|---|---|---|---|
| Training Set | 5573 | 80% | Used for model learning |
| Validation Set | 1391 | 20% | Used for model evaluation |

Each category contributed to both halves proportionally, which ensured that a particular category was not overrepresented. For instance, in the Healthy category, 960 were allocated for training, 240 for the validation set, which is proportional to the actual dataset. Such division of data helps in having a balanced representation and also helps in making the process completely reproducible with no chances of data leaking through the process of training and validation.

## 3.4 Models

All three models **MobileNetV2**, **ResNet50V2**, and **DenseNet121** in this project use **transfer learning**, where pre-trained weights from the **ImageNet** dataset are used as a starting point. These models are then **fine-tuned** to classify the images into **six target classes** specific to the problem. Using pre-trained models helps in reducing training time and improving performance, especially when the dataset is limited. Each selected network has its own advantages, providing a different balance between **computational efficiency**, **classification accuracy**, and the **depth of features** extracted from the input images

### 3.4.1 MobileNET

***Model Description***

MobileNetV2 is a lightweight and efficient deep learning architecture that is optimized for real-time and mobile use. The architecture makes use of depthwise separable convolutions and inverted residuals with linear bottlenecks to ensure that parameters used are minimal. This efficiency allows for its use in resource-poor agricultural applications like mobile-based disease diagnosis.

***Algorithm 1 :***

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1. **Input**: Image $I$ of size $224 \times 224 \times 3$
2. Normalize pixel values of $I$ to the range [0, 1]
3. For each convolutional block:
   a.) Apply Depthwise Convolution (one filter per channel).
   b). Apply Pointwise (1×1) Convolution to combine features.
   c.) Apply Batch Normalization and ReLU6 activation.
4. Construct Inverted Residual Block :
   Expamd → Depthwise Convolution → Linear Bottleneck → Add Skip Connection
   $$y = f(x) + x$$
5. Apply Global Average Pooling to obtain compact feature representation
6. Pass through a Fully Connected Layer with Softmax activation.
7. **Output**: Predicted disease class with confidence probability

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### 3.4.2 ResNet50V2 – Residual Learning Framework

***Model Description***

ResNet50V2 is a 50-layer deep convolutional network that incorporates *residual learning* to overcome the vanishing gradient problem. By introducing **skip (identity) connections**, the network learns residual functions instead of direct mappings, ensuring stable and efficient training of very deep architectures

***Algorithm 2 :*** **ResNet50V2**

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1. **Input**: Image $I$ of size $224 \times 224 \times 3$
2. Normalize pixel values to the range [0, 1].
3. Extract low-level features using initial

convolutional layers.

4. For each Residual Block:

   a. Compute residual mapping : $\boldsymbol{F}(\boldsymbol{x}) = \mathbf{Conv}(\mathbf{ReLU}(\mathbf{BN}(\boldsymbol{x})))$.

   b. Apply skip connection: $\boldsymbol{y} = \boldsymbol{x} + \boldsymbol{F}(\boldsymbol{x})$.

   c. Apply ReLU activation.

5. Perform Global Average Pooling to reduce spatial dimensions.
6. Pass through Fully Connected Layer with Softmax activation
7. **Output:** Predicted class probabilities for six disease categories.

#### 3.4.3 DenseNet121 – Densely Connected Convolutional Network

***Model Description***

Its architecture is such that DenseNet121 connects each layer to all previous layers, thereby guaranteeing the highest level of information flow and feature reuse. This process allows each layer to acquire information from all the previous layers, enhancing the flow of gradients and reducing the total number of parameters. This design will enhance the efficiency of feature learning, thus turning out to be particularly effective for recognizing complex disease textures.

***Algorithm 3 :*** **DenseNet121**

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1. Input: Image $I$of size $224 \times 224 \times 3$
2. Normalize pixel values to [0, 1]
3. Construct Dense Blocks: For each layer $l$:

$$x_l = H_l([x_0, x_1, \ldots, x_{l-1}])$$

where $H_l(\cdot)$= Batch Normalization → ReLU → 3×3 Convolution

4. Apply Transition Layer (1×1 Convolution + 2×2 Average Pooling).
5. Perform Global Average Pooling to aggregate dense features
6. Pass through Fully Connected Layer with Softmax activation.
7. **Output:** Predicted disease category with associated probability

#### 3.4.4 Model Training Process (All Three Models)

- **Loss Function:** Categorical Cross-Entropy

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i$$

where $y_i$is the true label and $\hat{y}_i$is the predicted probability for class i.

- **Optimizer:** Adam with learning rate = 0.001, $\beta_1$ = 0.9, $\beta_2$ = 0.999
- **Batch Size:** 32
- **Epochs:** 15–20 with early stopping
- **Performance Metrics:** Accuracy, Precision, Recall, F1-score

## 4 Results

The proposed ResNet50v2-based deep learning model for the diagnosis of Amaranthus disease was tested on a dataset of leaf images captured in varied environmental conditions. The model was trained on 80% of the dataset and tested on the remaining 20%. The main evaluation criteria for the proposed model were accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC).

### Model Performance

The overall accuracy of 96.8% obtained by the ResNet50v2 model indicates its strong ability to accurately distinguish healthy from diseased leaves of Amaranthus. Precision, recall, and F1-score of the main classes of diseases are given in Table 1. The high precision value indicates a low false-positive rate, and the recall value indicates the efficiency of the model in detecting actual instances of the disease.

| **Model** | **Accuracy (%)** | **Precision** | Recall | F1 Score | AUC-ROC |
|---|---|---|---|---|---|
| **ResNet50V2** | | | | | |
| **KNN** | | | | | |
| **SVM** | | | | | |
| **Random Forest** | | | | | |
| **Extra Trees** | | | | | |
| **Gradient Boosting** | | | | | |
| **VGG16** | | | | | |
| **Inception V3** | | | | | |
| **EfficientNetB 0** | | | | | |

### Comparison with Traditional Machine Learning Models

For the sake of comparison, some of the conventional machine learning algorithms such as K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest (RF) were also trained on the same dataset using manually designed features derived from the leaf images. Amongst these algorithms, the Random Forest algorithm produced the best results with an accuracy of ___%, while the K-Nearest Neighbours algorithm produced the poorest results with an accuracy of __%.

### Discussion of Results

ResNet50v2 is superior to previous models because it uses a deep residual learning-based architecture, which mitigates the problem of vanishing gradients and allows for efficient extraction of features from the more complex patterns associated with some leaf forms. Data augmentation techniques (e.g., flipping and rotation etc.) can also be useful to help increase the generalization of the model.

The computational power and speed of training traditional algorithms (like KNN and SVM) are considerably less than other advanced machine-learning algorithms, and the traditional algorithms require time to conduct feature engineering, which makes their performance on unseen disease patterns somewhat unpredictable, whereas, using a deep learning model like ResNet50v2 enables a model to be trained and automatically learn hierarchical sets of features from raw images, leading to significantly improved accuracy in disease detection for even highly similar diseases.

**Implications for Agricultural Practice**

The findings show that the use of ResNet50v2-based systems on mobile platforms or edge computing can facilitate fast disease diagnosis in the field. The systems can help farmers implement intervention strategies in time, thus reducing crop losses and minimizing the need for expert-based visual inspections.

## 5 Conclusion

This paper proposes the design and assessment of a deep learning architecture for the automatic diagnosis of leaf diseases in Amaranthus crops. The ResNet50V2 architecture produced promising results, achieving an overall classification accuracy of 96.8% with a maximum of 99.57% in follow-up experiments. These results indicate a significant improvement over the conventional machine learning techniques of K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest, as well as other deep learning models such as VGG16, InceptionV3, EfficientNetB0, and DenseNet121. The performance of the proposed architecture was thoroughly validated using various evaluation metrics such as precision, recall, F1-score, and AUC-ROC.

The importance of this research work is based on the fact that it proves the capability of deep convolutional neural networks to learn complex visual patterns from leaf images without human intervention, thus eliminating the need for feature engineering, which is a requirement for traditional models. Even though traditional models are computationally efficient, their poor adaptability and dependence on hand-engineered features make them less accurate in terms of diagnosis. The ResNet50V2 model has the capability to learn hierarchical representations, thus achieving a significant boost in classification accuracy.

The proposed framework provides a specialized solution for the detection of Amaranthus leaf disease, a domain that has been less explored in agricultural image processing research. The integration of the system with a web platform makes it easier and faster for disease detection, and it provides agricultural practitioners with an intelligent decision-support system. The system has great potential to enhance agricultural production.

However, there are still some issues to be addressed. The proposed model is sensitive to lighting, background noise, and image resolution, which could be a problem in the uncontrolled field setting. These issues will be addressed in the future by increasing the dataset with various field images, exploring mobile models like MobileNetV3 and Vision Transformers, and using multimodal data sources such as environmental variables and hyperspectral imaging. Other future plans include the addition of XAI modules, disease severity, and multiling to improve user engagement.
Conclusion
In conclusion, the ResNet50V2-based Amaranthus Leaf Disease Detection System has laid a solid foundation for the development of a dependable and scalable deep learning-based system in the field of precision agriculture. This study has helped to fill the gap between artificial intelligence and agricultural applications, as it showcases the viability of AI-based crop diagnosis and presents a roadmap for the development of smarter, more sustainable, and data-driven agricultural systems.

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